D06 · Logistics Technology & Innovation

# 03 · Autonomous delivery & drone logistics

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## 1 WHAT IT IS

Autonomous delivery covers drones and autonomous ground vehicles that move parcels without a driver on board, currently viable in specific niches such as medical samples, remote access and closed campuses.

## 2 WHY IT MATTERS

The technology works; the constraints are regulation, cost at scale, weather, airspace and public acceptance. **In Mexico, drone operations fall under AFAC rules (NOM-107-SCT3), which continue to evolve.** The opportunity today is where time or access is extremely valuable, not general e-commerce, so pilots must be judged on mature-state economics and regulatory feasibility, not on demonstrations.

## 3 HOW IT IS DONE

1. **Pick a niche where time or access is critical.** Medical samples, urgent spare parts, remote sites, closed campuses.
2. **Engage the regulator early.** Airspace approvals and operating conditions are on the critical path.
3. **Define success criteria before the pilot.** Delivery success rate, cost per delivery, safety incidents, customer value.
4. **Compare with the best conventional alternative.** Fully loaded cost and service versus courier or ground transport on the same use case.

### IN PRACTICE

**Focus pilots on niches where the value of time or access is extreme**

General e-commerce last-mile is still a difficult economic and regulatory environment for autonomy.

**Engage regulators early**

In Mexico the AFAC framework is a critical path item. Technical feasibility without regulatory permission has no operational value.

**Measure real TCO in the pilot, not brochure numbers**

Pilot-phase costs are higher; the question is whether the mature-state economics and service benefits justify the journey.

### ILLUSTRATIVE CASE

*Illustrative case built from typical industry values — not data from a specific company.*

**Illustrative case — Hospital network evaluating drones for urgent lab samples (CDMX area)**

| Criterion | Ground Courier | Drone (proposed pilot) |
|---|---|---|
| Average urgent delivery time | 47 min (highly variable) | 12–18 min (weather permitting) |
| Cost per delivery | $85 average | ~$40 mature / ~$120 pilot phase |
| Reliability | Traffic dependent | Weather and airspace dependent |
| Regulatory readiness | Existing | Requires specific AFAC corridor approval |

**Result:** Pilot approved for a limited high-urgency corridor after regulatory engagement. The business case is positive for the medical-sample use case even at pilot-phase costs because of the clinical value of time.

### HOW IT IS MEASURED

**Drone / ADV Delivery Success Rate %**

(Successful autonomous deliveries / Attempted autonomous deliveries) × 100

*Benchmark: >95% required for operational credibility; >99% expected in regulated medical use cases*

**Autonomous vs Traditional Cost per Delivery**

Fully loaded cost comparison on the same lanes or use cases

*Benchmark: Must reach clear advantage or strategic necessity to justify scale investment*

#### WHAT YOU WOULD USE

**Autonomous Delivery**

**Wing (Alphabet) / Zipline / other specialized drone operators** — Most mature commercial drone delivery platforms, especially in medical and remote applications.

**Starship and similar ground ADV platforms** — Relevant for campus, closed-community and certain urban sidewalk/road applications.

## 4 SO WHAT

**Pilot where the value of time is extreme.** Engage AFAC early, set success criteria upfront and compare honestly with the best ground alternative before scaling.

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